

Literature-implied answer for arXiv:1001.2026

Spanning unimodular eigenvectors need not be perfectly spanning

Status. literature_implied_answer. This is not a new proof produced by the run. It is a direct identification of a later counterexample theorem with one of the open questions in Sophie Grivaux, *A new class of frequently hypercyclic operators*, arXiv:1001.2026.

Source question. In Section 6, “Open questions and remarks”, subsection “Hypercyclic operators with spanning unimodular eigenvectors”, Grivaux asks:

If T is a hypercyclic operator on X whose eigenvectors associated to eigenvalues of modulus 1 span a dense subspace of X , is it true that the unimodular eigenvectors of T are perfectly spanning?

This is Question 2 in the source paper’s notation.

Supporting literature. Quentin Menet, *Linear chaos and frequent hypercyclicity*, arXiv:1410.7173, Theorem 1.2, constructs a chaotic operator T on ℓ^1 which is neither $\mathcal{U}$ -frequently hypercyclic nor distributionally chaotic; in particular, T is chaotic and not frequently hypercyclic. The same introduction recalls the standard identity, for complex vector spaces, that periodic points are spanned by eigenvectors associated to roots of unity.

Identification. Menet’s operator is chaotic, so it is hypercyclic and has a dense set of periodic points. Since periodic points lie in the span of root-of-unity eigenvectors, the eigenvectors associated to eigenvalues of modulus 1 span a dense subspace of ℓ^1 . If these unimodular eigenvectors were perfectly spanning, Grivaux’s main Theorem 1.1 in arXiv:1001.2026 would imply that T is frequently hypercyclic. This contradicts Menet’s Theorem 1.2. Therefore Menet’s operator is a negative example to Grivaux’s Question 2.

Claim. *There exists a bounded hypercyclic operator on a complex Banach space whose unimodular eigenvectors span a dense subspace, but whose unimodular eigenvectors are not perfectly spanning.*

Scope and limitations. The chaotic-implies-frequently-hypercyclic question quoted immediately before Grivaux’s Question 2 is explicitly answered negatively by Menet and is already recorded in this run in the packet 1005.1416_chaotic_not_fhc_answered_by_1410.7173. The present packet records the additional implied answer to Grivaux’s “perfectly spanning” question. This packet does not answer the related question in Grivaux’s final section asking for an operator whose full unimodular point spectrum is countable and whose corresponding eigenspaces span densely. The argument above uses only the dense root-of-unity eigenspaces forced by dense periodic points, and does not verify the full unimodular point spectrum of Menet’s operator.

Search evidence. The run’s cheap indexes had no direct packet for arXiv:1001.2026 or this perfectly-spanning implication. They did contain the adjacent arXiv:1005.1416/arXiv:1410.7173 chaotic-not-frequently-hypercyclic packet. Local source search and web/arXiv search were then used for the key phrases “spanning unimodular eigenvectors”, “perfectly spanning”, and “chaotic operator frequently hypercyclic”.

References

- [1] S. Grivaux, *A new class of frequently hypercyclic operators*, arXiv:1001.2026; Indiana Univ. Math. J. 60 (2011), 1177–1201.
- [2] Q. Menet, *Linear chaos and frequent hypercyclicity*, arXiv:1410.7173.